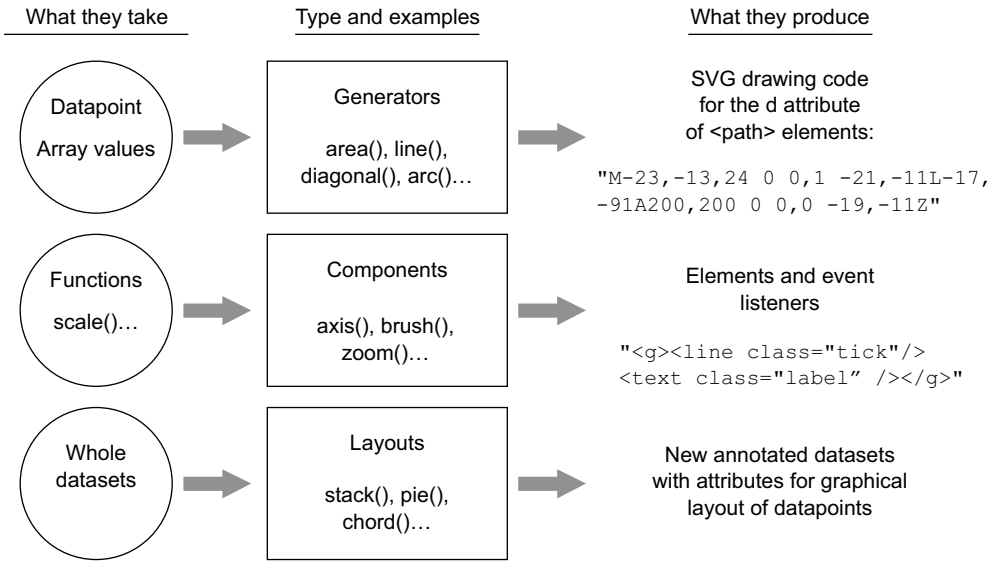


Figure 4.2 The three main types of functions found in D3 can be classified as generators, components, and layouts. You'll see components and generators in this chapter and layouts in the next chapter.

4.1.1 Generators

D3 generators consist of functions that take data and return the necessary SVG drawing code to create a graphical object based on that data. For instance, if you have an array of points and you want to draw a line from one point to another, or turn it into a polygon or an area, a few D3 functions can help you with this process. These generators simplify the process of creating a complex SVG `<path>` by abstracting the process needed to write a `<path>` `d` attribute. In this chapter, we'll look at `d3.svg.line` and `d3.svg.area`, and in the next chapter you'll see `d3.svg.arc`, which is used to create the pie pieces of pie charts. Another generator that you'll see in chapter 5 is `d3.svg.diagonal`, used for drawing curved connecting lines in dendrograms.

4.1.2 Components

In contrast with generators, which produce the `d` attribute string necessary for a `<path>` element, components create an entire set of graphical objects necessary for a particular chart component. The most commonly used D3 component (which you'll see in this chapter) is `d3.svg.axis`, which creates a bunch of `<line>`, `<path>`, `<g>`, and `<text>` elements that are needed for an axis based on the scale and settings you provide the function. Another component is `d3.svg.brush` (which you'll see later), which creates all the graphical elements necessary for a brush selector.

4.1.3 Layouts

In contrast to generators and components, D3 layouts can be rather straightforward, like the pie chart layout, or complex, like a force-directed network layout. Layouts